MILITARY SPECIFICATION SHEET

# RELAYS, SOLID STATE, TIME DELAY (ON OPERATE), TRACK MOUNTED, TYPE I, CLASS C, SPST, 150 MILLIAMPERES, FIXED TIME, 50 MILLISECONDS TO 500 SECONDS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83726.

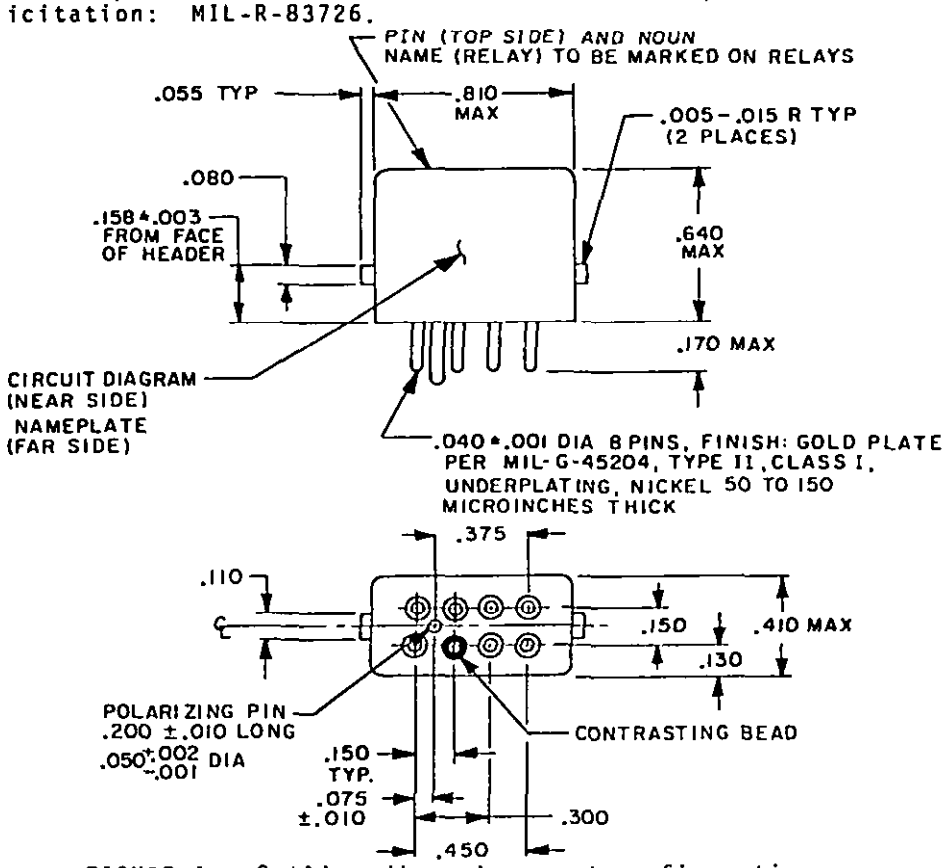
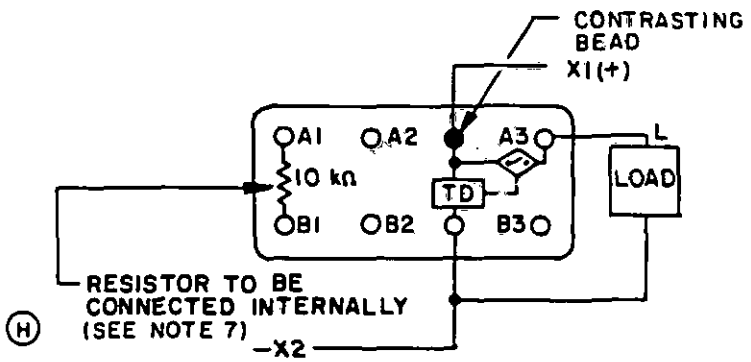



FIGURE 1. Outline dimensions and configuration.

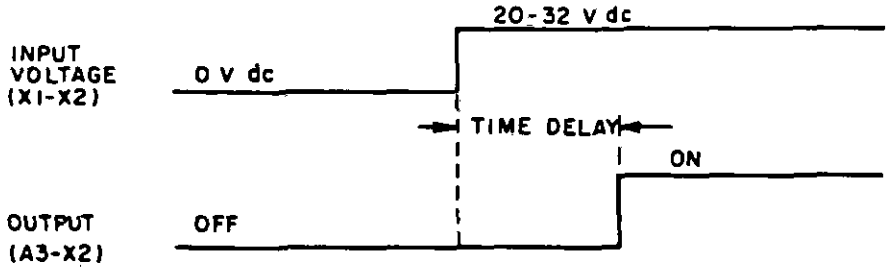
Ⓗ denotes changes


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CIRCUIT DIAGRAM
(SEE NOTES 4 AND 5)




TIMING DIAGRAM

| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .110 | 2.79 |
| .002 | 0.05 | .130 | 3.30 |
| .003 | 0.08 | .150 | 3.81 |
| .005 | 0.13 | .158 | 4.01 |
| .010 | 0.25 | .170 | 4.32 |
| .015 | 0.38 | .200 | 5.08 |
| .040 | 1.02 | .300 | 7.62 |
| .050 | 1.27 | .375 | 9.52 |
| .055 | 1.40 | .410 | 10.41 |
| .075 | 1.90 | .450 | 11.43 |
| .080 | 2.03 | .640 | 16.26 |
| | | .810 | 20.57 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
(H) 4. Terminals A1, A2, B1, and B2 are connected internally. Terminals A1 and B1 can be used externally. Terminals A2 and B2 shall not be used for external tie points or for terminals.
5. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
6. Relays are intended for use with relay socket mount MIL-R-6106/23-001 and relay bracket mount MIL-R-6106/25.
(H) 7. The 10 kΩ resistor shall be rated at 250 milliwatts minimum.

FIGURE 1. Outline dimensions and configuration - Continued.



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REQUIREMENTS:

OPERATING DATA:

Timing action: Delay-on operate.

Time delay: Fixed; select from .05 to 500 seconds.

Timing accuracy 1/: ±10 percent of the nominal value.

Recycle time 2/: 10 milliseconds.

Power interrupt: The accuracy will not be affected by power interruptions up to 1 millisecond spaced at least 10 milliseconds apart.

INPUT DATA:

Input voltage: 28 V dc nominal; range 20 to 32 V dc.

Duty rating: Continuous.

Current drain: 10 milliamperes maximum plus load at +25°C.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

OUTPUT DATA:

Configuration: SPST; switch closure to +28 V dc.

Rating: 150 milliamperes maximum.

Suppression: Inductive suppression provided for output protection.

Voltage drop: 2 V dc maximum.

Leakage current: 1.0 microampere maximum at 28 V dc and +25°C; 10 microamperes maximum at 28 V dc and +125°C.

Endurance (life): 1,000,000 cycles minimum for test.

Qualification: 2,000 hours.

Group B: 1,000 hours or 1,000,000 cycles, whichever is less.

ELECTRICAL DATA:

Transients 3/:

Positive transients: +80 volts in accordance with MIL-STD-704, figure 9, limit 1.

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1/ *The accuracy requirement applies for any combination of operating temperature and voltage.*

2/ Recycle time is defined as the minimum time that power must be removed from the input terminals to assure that the next timing cycle will be completed within the specified timing tolerance. (Units can be recycled during timing or after time-out.)

3/ Transient specifications are based on a maximum duty cycle of 1/50.



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Self-generated transients: ±50 volts maximum.

Spike susceptibility: ±600 volts, 10 microseconds maximum.

Electromagnetic interference 4/: In accordance with MIL-STD-461, class 1D.

Insulation resistance 5/: 1,000 megohms at 500 V dc between each pin and case.

Dielectric strength 5/: 1,000 volts rms at 60 hertz at sea level between case and pins connected together.

ENVIRONMENTAL DATA:

Ambient temperature (operating and nonoperating): -55°C to +125°C.

Vibration (sinusoidal): 10 to 55 hertz at 0.06 inch DA; 10 g, 55 to 500 hertz; 20 g, 500 to 3,000 hertz.

Vibration (random): MIL-STD-202, method 214, condition I, letter C, duration of 30 minutes.

Shock: 1,100 g for 0.5 millisecond, 1/2 sine, 3 axis.

Acceleration: 100 g, 3 axis.

Seal: Hermetic.

Moisture resistance: MIL-STD-202, method 106.

PHYSICAL DATA:

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 0.56 ounce.

Marking: Part or Identifying Number (PIN), circuit diagram, manufacturer's part number and name, and date code.

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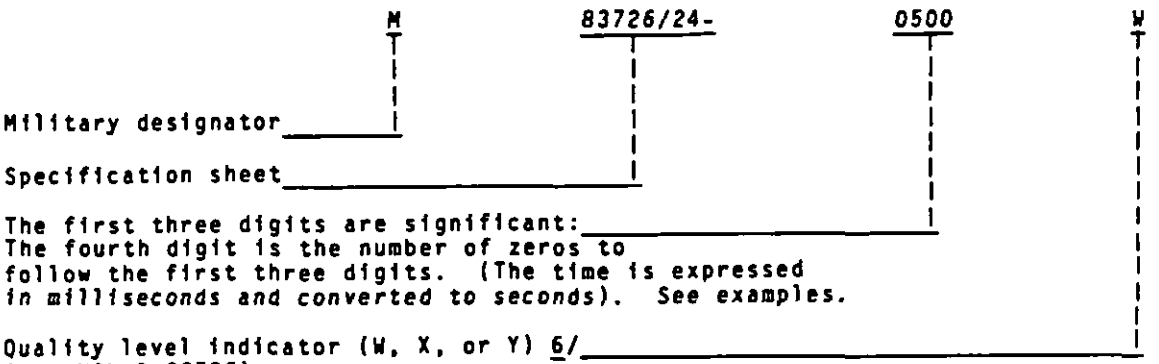
4/ EMI test limits will not be exceeded during the timing interval or when continuously energized under steady-state conditions per radio interference requirements of MIL-R-83726.

5/ Terminals X1, X2, and A1, A3, and B1 must be connected together during this test. Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and between all terminals and case.



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PIN: Consists of the letter M, basic number of this specification sheet, a four digit dash number (expressed in milliseconds), and a quality level indicator as follows:




Examples: M83726/24-0500W 50 millisecond time delay W level.
M83726/24-2503X 250 second time delay X level.
M83726/24-5003Y 500 second time delay Y level.

6/ Any relays numbered prior to the date of this specification without a quality level indicator shall be considered interchangeable (store and issue) with the "W" quality level.

## CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F699)



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